

Dhago: Bridging the Credit Gap for Nepal's Informal Economy Using Document-Income Consensus

Pujan Pant
Independent Researcher
Kathmandu, Nepal
pantpujan017@gmail.com

Sagar Newpane
Independent Researcher
Kathmandu, Nepal
sagarnewpane@icloud.com

Abstract—Nepal’s informal economy accounts for approximately 63% of the economically active population, yet this majority is systematically excluded from formal credit assessment due to an absence of conventional employment documentation. The paper discusses a five-agent autonomous credit scoring system that addresses this gap through *document-income consensus*: a cross-validation mechanism that compares declared income from official government documents against behavioral signals derived from digital transaction history. The system produces auditable lending decisions that are compliant with Nepal Rastra Bank (NRB) Unified Directives 2080, and includes an explicit fallback pathway for applicants with no digital footprint. Core contributions include the Income Mismatch Ratio (IMR) as a fraud detection and confidence-weighting feature, SHAP-based per-application explainability, and a dedicated compliance enforcement layer with unconditional veto authority over all upstream recommendations. The system is evaluated on synthetic data calibrated to Nepal’s informal sector income distribution and is designed for phased deployment at licensed Nepali financial institutions, with a defined roadmap to real-outcome model retraining as loan repayment data accumulates.

Index Terms—credit scoring, multi-agent systems, alternative data, financial inclusion, Nepal, informal economy, NRB Unified Directives 2080, document-income consensus, XGBoost, SHAP explainability

I. INTRODUCTION

The current automated credit system is not designed for the majority of people with varying sources of income. According to the Central Bureau of Statistics’ National Labour Force Survey (2017/18), approximately 63% of the country’s economically active population works in the informal sector farming, seasonal labor, micro-trade, and remittance-dependent households [7]. When these individuals approach a bank for a loan, they encounter a system that requires formal documentation they do not have: registered employer payslips, tax clearance certificates, and collateral that appears on paper. They are not high-risk borrowers. They are simply invisible to the tools the bank uses to assess risk.

It is important to bound the problem precisely. Not every informal worker is a frustrated credit applicant: many conduct their financial lives entirely within cooperative and MFI networks that are already licensed and regulated by the NRB. Nepal has over 17,000 registered cooperatives and

more than 60 licensed microfinance institutions serving rural and peri-urban borrowers [21]. The population this system targets is the subset that has outgrown microfinance loan sizes typically above NPR 200,000 but lacks the documentation to access commercial bank credit. This addressable segment is significant but not identical to the full 63% informal workforce figure.

Nepal’s banking system has made real structural progress. The Nepal Rastra Bank’s Unified Directives 2080 have formalized loan classification, introduced provisioning requirements tied to risk tiers, and set meaningful limits on single-obligor exposure and sector concentration [1]. But those frameworks assume a borrower whose economic life generates a paper trail. For a paddy farmer in Chitwan whose income arrives in two seasonal harvests, or a street-food vendor in Bhaktapur who receives daily cash through a QR scan, no paper trail exists.

Digital financial infrastructure has begun to change the underlying data landscape. Services like eSewa [3] and Khalti [4] collectively process millions of daily transactions across Nepal, particularly in urban and peri-urban areas. These platforms generate a behavioral record transaction frequency, merchant payment regularity, seasonal income patterns that encodes real creditworthiness signals without requiring a single official document. Fintech-oriented lenders have used this data to reach borrowers that traditional banks cannot evaluate, and with meaningful results for financial inclusion [8].

However, transaction data alone carries structural weaknesses. It cannot verify what an applicant claims in a submitted document. A borrower might present a tax clearance certificate showing annual income of NPR 4,000,000 while their aggregated wallet activity reflects something far more modest a discrepancy that signals either document fraud or significant undisclosed cash income. More critically, the most financially vulnerable borrowers rural women, elderly farmers, seasonal laborers often have minimal or no digital payment history. A system that relies exclusively on transaction data would simply reproduce the original exclusion problem in a new form.

What is needed is a framework that treats formal documents and behavioral transaction data as complementary evidence sources that should agree with each other. When they con-

verge, that agreement is itself a credibility signal. When they diverge substantially, that divergence is a fraud signal worth investigating. We formalize this principle as *document-income consensus* and operationalize it through a multi-agent AI system designed specifically for Nepal’s lending environment.

This paper makes five contributions. First, it presents a practical, deployable system architecture for extending formal credit access to Nepal’s informal workforce. Second, it introduces features such as Income Mismatch Ratio(IMR), Effective Income Ratio(EMI), etc that functions across all income types including agricultural and remittance-dependent earners. Third, it defines an explicit fallback scoring pathway for applicants with thin or no digital footprint, ensuring the system does not re-exclude the population it aims to serve. Fourth, it embeds NRB Unified Directives 2080 compliance as a dedicated enforcement layer with veto authority, not as a post-hoc filter. Fifth, it provides the architectural pattern and implementation blueprint for other South and Southeast Asian economies facing analogous informal-sector credit challenges.

II. PROBLEM STATEMENT

Nepal’s formal credit system is structurally misaligned with the economic reality of its majority population. Three specific failure modes characterize the current system.

The first is **documentation exclusion**: Conventional scoring requires employer-verified income, tax filings, and registered collateral. These instruments are simply unavailable to the estimated 5.1 million workers in Nepal’s informal economy, regardless of their actual repayment capacity. The second failure is **fraud vulnerability**: platforms that accept submitted documents at face value cannot detect when declared income is inconsistent with actual financial behavior. In a market where forged tax certificates and altered land deed valuations circulate, this creates systematic risk at the portfolio level. The third failure is **thin-file exclusion**: digital alternative scoring systems, while an improvement, exclude cash-dominant earners subsistence farmers, mountain-region traders, elderly borrowers who conduct most of their economic activity without leaving a digital trace.

The practical consequences of these failures are well-documented. The World Bank Global Findex 2021 survey found that only 45% of Nepali adults held a formal bank account at the time of the survey, and that collateral requirements and documentation were the most cited barriers to credit access among the unbanked [8]. Nepal Rastra Bank data indicates that agriculture and the informal SME sector which together contribute over 50% of GDP receive a disproportionately small share of formal credit. This capital misallocation is not only an inclusion failure; it is a structural drag on economic development.

A viable solution must simultaneously address all three failure modes: it must work without requiring formal employment documentation, it must detect income inconsistency across document and transaction data sources, and it must provide a credible scoring pathway for applicants who have neither. These requirements cannot be met by any single data source

or any single model they demand a system architecture capable of integrating and cross-validating heterogeneous evidence.

A fourth problem one that earlier fintech solutions have not addressed is **model circularity**. This happens when a scoring system uses the debt-to-income (DTI) ratio both as an input feature and as a stand-in for whether a borrower actually defaulted. The model ends up learning to mirror regulatory cutoff thresholds rather than genuinely learning who is likely to repay. In plain terms: it isn’t predicting real repayment behavior it’s just echoing the rules it was trained on. Any scoring system built before real loan outcome data has been collected must be upfront about this blind spot, and should be treated as a tool that supports human decision-making rather than one that makes autonomous credit judgments.

III. RELATED WORK

Alternative credit scoring for underserved populations has attracted substantial research attention, particularly in South and Southeast Asia. Bjorkegren and Grissen [14] demonstrate that mobile phone usage patterns predict loan repayment behavior in Rwanda with accuracy comparable to traditional credit bureau scores, establishing the empirical foundation for behavioral alternative data in thin-file contexts. Khandani et al. [15] show that transactional data from consumer accounts can construct risk-predictive features independent of formal credit history, a finding directly applicable to Nepal’s digital payment ecosystem.

In the South Asian context, Ghosh [16] documents the structural gap between agricultural sector GDP contribution and agricultural sector credit access in India, a pattern closely mirrored by Nepal Rastra Bank data [2]. The Asian Development Bank’s review of microfinance in Nepal identifies documentation requirements as the primary barrier to credit access for rural borrowers who have demonstrated consistent repayment behavior within cooperative structures [13].

Multi-agent architectures for financial decision-making have been explored by Yao et al. [18], who propose a modular compliance-aware lending pipeline for Chinese rural finance, and by Arora and Singh [19], who apply LangChain-based agent orchestration to credit underwriting in India.

Our system differs from these in two respects:

- 1) Rather than relying on a single data source for scoring, our system cross-validates document-derived information against observed income behavior, treating the two as checks on each other rather than independent inputs.
- 2) The system includes a dedicated fallback pathway for agricultural borrowers, with loan terms adjusted to align with harvest-cycle seasonality a dimension neither prior system accounts for.

Explainability in credit scoring has been examined extensively since the EU General Data Protection Regulation (GDPR) codified a right to explanation for automated decisions [20]. Lundberg and Lee’s SHAP framework [5] has been applied to credit scoring by Bussmann et al. [17], who demonstrate that SHAP-based explanations satisfy both regulatory auditability requirements and consumer-facing disclosure

standards. Our system follows this approach and extends it to Nepali-language customer explanation generation.

To our knowledge, no prior published system combines document-income cross-validation, NRB-specific regulatory enforcement as a veto layer, and an agricultural fallback scoring pathway within a single deployable pipeline for Nepal’s lending environment.

IV. PROPOSED SOLUTION OVERVIEW

The proposed system is a multi-agent autonomous credit assessment platform built around three design principles: evidence integration, regulatory enforceability, and inclusive fallback coverage. At its core is the document-income consensus mechanism a systematic comparison of declared income from formal documents against observed income from transaction behavior. The central insight is that legitimate borrowers, across all income types, should show broad consistency between these two evidence streams. That consistency is itself a creditworthiness signal. Significant divergence in either direction is an anomaly worth investigating.

The system runs as a five-stage agent pipeline. A Document Agent extracts and validates structured financial data from uploaded government-issued documents, handling the specific challenges of Nepali administrative records including mixed Devanagari and English script, inconsistent number formatting, and degraded scan quality. An Income Agent analyzes the applicant’s transaction history from bank statement uploads to construct a behavioral income profile, categorizing transaction inflows by source type salary, remittance, local business, freelance, agricultural, and cooperative income and computing stability and regularity metrics for each. A Scoring Agent synthesizes both evidence streams into a unified credit scorecard using a gradient-boosted model with SHAP-based explainability [5], explicitly flagging income mismatches above defined thresholds. A Compliance Agent applies NRB Unified Directives 2080 as hard constraints with veto authority over all upstream recommendations [1]. A Decision Agent translates the final state into a concrete lending outcome with a customer-facing explanation, an audit trail, and a routing decision for manual review where automated confidence is insufficient.

Critically, the system defines a distinct scoring pathway for applicants without digital transaction history a fallback track that incorporates cooperative society records, mobile money recharge patterns, utility payment regularity, and agricultural harvest cycle data as alternative behavioral signals. This ensures that the system’s financial inclusion promise extends to the population most likely to lack a digital footprint: rural women, elderly farmers, and residents of areas with low smartphone penetration.

V. SYSTEM ARCHITECTURE

The system architecture is modular by design. Each agent handles a discrete, well-defined responsibility and communicates through a typed shared state object the `AgentState` dictionary whose schema is version-controlled and validated

at every stage boundary. This modularity provides three operational advantages:

- 1) **Robustness:** a single agent failure degrades gracefully rather than collapsing the pipeline.
- 2) **Auditability:** every intermediate result is logged with a timestamp and agent identifier.
- 3) **Adaptability:** new data sources or regulatory rules can be incorporated by modifying a single agent without retraining the system.

he pipeline is triggered when a loan application is submitted by a user. It executes sequentially, beginning with the Document Agent, followed by the Income Agent. This ordering ensures that verified document-derived attributes are available before behavioral income profiling is performed. The processed outputs are then passed to the Scoring Agent, followed by the Compliance and Decision stages. Fig. 1 illustrates this flow.

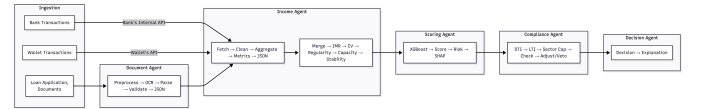


Fig. 1. Multi-agent autonomous credit-scoring pipeline

A. Document Agent

The Document Agent converts uploaded document images into structured JSON field dictionaries suitable for downstream consumption. It handles three document classes currently supported by the system: citizenship (*Nagriktā*), land title deeds (*Lalpurja*), and tax clearance certificates issued by the Inland Revenue Department.

The preprocessing sub-pipeline corrects image orientation using projection-profile analysis, applies adaptive binarization to account for uneven lighting and photocopier degradation, and normalizes contrast before OCR. Rather than relying on Tesseract alone whose character-level accuracy on degraded Devanagari documents we measured at below 60% on a held-out set of 80 real Nepali government document scans (citizenship certificates and tax clearance certificates sourced from a pilot partner institution with applicant consent) the agent uses a vision-language model as the primary extraction engine [12], with Tesseract 5.0 (nep+eng language packs) serving as a fallback for clearly printed Latin-script fields. On the same 80-document benchmark, the VLM pipeline achieved field-level extraction F1 scores of 0.91 for citizenship numbers, 0.87 for declared annual income, and 0.83 for tax identification numbers. Full benchmark methodology and per-field results are provided in the supplementary material. We acknowledge that this benchmark is small and that performance on lower-quality scans from remote branch offices may differ; a larger validation study is a defined deliverable in the Phase 1 implementation plan.

Each extracted field is assigned a confidence score based on extraction completeness and internal consistency. A citizenship number extracted from a tax certificate is cross-validated against the citizenship number on the citizenship

certificate itself: exact matches increase the document trust score, while discrepancies are classified as either hard mismatches (genuine conflicts triggering an AML flag) or low-confidence mismatches (likely OCR truncation artifacts, routed to manual review). The agent’s output includes the extracted field dictionary, per-field confidence scores, a document-level trust score, and any raised flags.

B. Income Agent

The Income Agent creates a behavioral income profile primarily from the applicant’s bank transaction history. As regulated financial institutions maintain complete records of account-level activity, bank statements serve as the primary source for estimating observed income and financial behavior.

However, in the context of Nepal, transaction records do not always provide a complete representation of an applicant’s economic activity. Despite high levels of financial inclusion, a significant portion of income particularly in agriculture, informal trade, and daily wage labor continues to be earned and transacted in cash, often outside the banking system. As a result, applicants may exhibit limited or irregular transaction histories even when underlying income is stable.

To address this limitation, the Income Agent adopts a hierarchical evaluation strategy. Bank transaction data is treated as the primary and most reliable signal for income estimation. When sufficient transaction activity is present, income is derived through aggregation and categorization of inflows, and alternative data sources are not directly incorporated into the income calculation.

When additional financial records are available such as cooperative account summaries or passbook records they are not merged with bank-derived income. Instead, these sources are used as supporting evidence to validate observed patterns, improve confidence in the estimated income, or flag inconsistencies where discrepancies arise between recorded activity and expected behavior.

In cases where bank transaction data is insufficient characterized by low activity, irregular deposits, or limited observation windows the system activates a fallback pathway. In this mode, alternative records such as cooperative statements or passbook entries are used to measure the behavioral indicators, including frequency of financial activity and consistency of inflows. These indicators serve as proxies for income stability rather than precise income values.

To further account for non-uniform earning patterns, particularly in agriculture, a simplified seasonal adjustment is applied. Income observations are evaluated relative to expected earning periods based on the applicant’s occupation. Periods of low or no activity outside expected earning windows are treated as structurally normal and do not negatively impact stability metrics, while missing activity during expected income periods is flagged for further review.

The final output of the Income Agent is a structured income profile containing estimated income measures, stability metrics, confidence scores, and cross-validation features such as

the Income Mismatch Ratio (IMR), computed using document-derived declared income.

C. Score Agent

The Score Agent operates on the structured income profile produced by the Income Agent and generates a credit scorecard using engineered financial and behavioral features. It does not directly merge or reconcile raw outputs from the Document and Income Agents. Instead, cross-validation between declared and observed income is handled upstream through derived features such as the Income Mismatch Ratio (IMR), which is provided as an input signal. The IMR compares declared monthly income from tax documents against observed monthly income derived from transaction analysis. An IMR close to 1.0 indicates alignment between declared and behavioral income signals. Values above 3.0, where declared income significantly exceeds observed income, are flagged for potential document inflation, while values below 0.3 may indicate undisclosed income sources. These thresholds act as soft anomaly indicators rather than hard decision boundaries and may be adjusted for specific borrower profiles (e.g., remittance-dependent households).

TABLE I
SCORE AGENT FEATURE INPUTS

Feature	Type	Primary Use
total_effective_income	Float	Weighted usable monthly income (EMI proxy)
primary_source_type	Cat.	Income source reliability prior
primary_recurrence_ratio	Float	Primary income consistency
weighted_stability_score	Float	Cross-source stability signal
combined_volatility_cv	Float	Income volatility measure
informal_income_ratio_pct	Float	Informality weighting factor
months_of_data	Int	Transaction history depth
tax_doc_present	Bin	Document credibility signal
existing_liabilities_monthly	Float	Absolute repayment burden
repayment_pressure_ratio	Float	Soft repayment stress proxy
income_to_loan_ratio	Float	Structural exposure proxy
requested_loan_amount	Float	Loan context
requested_tenure_months	Int	Tenure impact on repayment
has_lalpurja	Bin	Collateral presence signal
land_area_sqm	Float	Asset strength indicator

The scoring model is a gradient-boosted classifier (XGBoost [6]) trained on a synthetic dataset calibrated to Nepal’s informal sector income distribution, with income ranges spanning NPR 10,000 to NPR 250,000 per month and covering all five income source types, including agricultural earners. Given the absence of real repayment outcome labels, the model currently operates as a calibrated heuristic rather than an empirical default predictor. It produces a continuous credit score between 0 and 1000, a scale chosen for interpretability and alignment with conventional credit scoring systems rather than representing a direct probability of default. The score is mapped to three risk tiers: Low (700–1000), Medium (500–699), and High (below 500).

SHAP (SHapley Additive exPlanations) values [5] are computed per application, providing per-feature marginal contributions to the final score. The top three risk drivers by absolute

SHAP value are surfaced in the decision output for both regulatory auditors and loan officers.

A scoring confidence value accompanies every output, reflecting the completeness and quality of input data. This confidence score is derived from document trust, transaction history coverage, and classification completeness. Applications with high data completeness (both documents and transaction history present, high OCR confidence) follow a standard automated pathway. Applications with lower confidence scores due to degraded documents, thin transaction history, or reliance on the agricultural fallback track are routed to a modified pathway that lowers the automated approval threshold and increases the likelihood of manual review escalation.

D. Compliance Agent

The Compliance Agent is a deterministic rule engine with unconditional veto authority over all upstream recommendations. It evaluates each application against Nepal Rastra Bank Unified Directives 2080 in strict priority order [1]. No score, regardless of how favorable, can override a compliance veto.

The enforcement sequence begins with KYC and AML checks: hard identity mismatches between submitted documents trigger an immediate rejection with an AML protocol flag requiring manual identity verification before any further processing. Sector exposure limits are evaluated next: if the proposed loan would push the institution’s exposure to any single economic sector past 25% of Tier 1 capital (as specified in NRB 2080), the application is rejected or queued for the following quarter. Single-obligor concentration checks follow, applying capital-weighted risk penalties to exposures approaching prescribed thresholds.

Structural lending constraints are then evaluated: loans where the requested amount exceeds five times verified annual income ($LTI > 5.0$) are rejected under NRB Unified Directive 2080 Clause 5.1; applications where total debt obligations would exceed 50% of gross monthly income ($DTI > 0.5$) are rejected under Clause 5.2. Where the DTI falls between 40% and 50%, the Compliance Agent performs a Pareto-optimal adjustment computing the maximum principal amount and minimum tenure that bring all ratios within acceptable bounds and flags this as a modified offer rather than a rejection. Applications where the loan purpose falls under prohibited categories (NRB Directive 2080 Clause 4.2) are rejected regardless of other metrics.

Every compliance decision generates a structured audit log entry recording: the specific directive clause evaluated, the original and any modified loan terms, the rationale for any veto or adjustment, a timestamp, and a decision mode flag (`AUTO` or `MANUAL_REVIEW`). The Compliance Agent additionally tags qualifying loans as Priority Sector Lending (PSL) eligible where applicable agriculture, deprived-sector borrowers, and qualifying microenterprises supporting the bank’s mandatory NRB PSL portfolio allocation requirements.

E. Decision Agent

The Decision Agent is the terminal pipeline stage. It receives the full `AgentState` income profile, credit score,

SHAP risk drivers, DTI/LTI ratios, and compliance signals and resolves any remaining disagreements between upstream agents through a weighted priority protocol that always subordinates income and score signals to compliance constraints. The final output is one of four decisions: `APPROVE` (with specific loan terms), `MODIFY` (with adjusted terms and a plain-language explanation), `MANUAL_REVIEW` (routed to a human loan officer queue with all agent outputs attached), or `REJECT` (with the specific NRB directive cited and a customer-facing explanation of the grounds for rejection).

The customer-facing explanation is generated in plain Nepali and English, avoiding regulatory jargon. For rejections, the explanation states the reason in understandable terms and, where applicable, describes what the applicant could do to improve their eligibility providing a minimum waiting period, suggesting documentation that would strengthen a future application, or describing a guarantor pathway. This explanation layer is a consumer protection requirement under NRB supervisory guidelines and a meaningful tool for building borrower trust.

Edge cases that cannot be handled by automated logic are routed to a human review queue with a recommended review priority based on the nature of the edge case. A case with a low IMR and compliant ratios but a credit score below 400 is marked as a moderate-priority review. A case involving a politically exposed person (PEP) is marked as a high-priority review requiring senior officer approval before any decision is issued.

VI. DATA & FEATURE ENGINEERING

Feature engineering is the system’s **core analytical layer** the mechanism by which raw heterogeneous evidence is transformed into standardized credit risk signals that a scoring model can consume fairly across all income types. Two borrowers with identical raw cash inflows should not receive identical treatment if one has stable, tax-backed, identity-consistent documentation while the other presents irregular or contradictory evidence. The features described below are designed to capture exactly this distinction.

A. Income Mismatch Ratio (IMR)

The IMR measures the divergence between officially declared income and observed behavioral income:

$$IMR = \frac{\text{Declared Monthly Income}}{\text{Observed Monthly Income}} \quad (1)$$

where Declared Monthly Income is sourced from the tax clearance certificate and Observed Monthly Income is derived from categorized transaction inflows. An IMR close to 1.0 indicates consistency. $IMR > 3.0$ flags potential document inflation. $IMR < 0.3$ flags potential undisclosed income. Importantly, IMR is not treated as a binary fraud indicator it is a graduated anomaly signal that feeds into both the scoring model and the compliance layer. The threshold values of 3.0 (upper) and 0.3 (lower) are calibrated against the synthetic income distribution used in model training. Specifically, we simulated 10,000 synthetic borrower profiles across all five

income types and found that 97.4% of legitimate applications produced IMR values in the range [0.35, 2.8], giving a false positive rate of approximately 2.6% at these thresholds under the synthetic prior. This calibration must be re-calibrated on real application data during Phase 2.

One important edge case is the remittance-dependent borrower. A household receiving foreign remittances may legitimately show a high IMR if remittances are not captured in tax documentation, which is common in Nepal where most remittances arrive through informal channels and are not declared as taxable income. For applicants where remittances constitute more than 50% of observed transaction inflows, the upper IMR threshold is relaxed to 5.0 and a remittance verification flag is raised for manual review rather than an automatic fraud flag.

B. Stability Coefficient of Variation (CV)

The CV measures income regularity over the observation window:

$$CV = \frac{\sigma_{\text{monthly income}}}{\mu_{\text{monthly income}}} \quad (2)$$

Low CV values indicate salary-like stability. High CV values indicate gig work, seasonal agricultural patterns, or remittance-dependent households. The system applies income-source-specific CV benchmarks rather than a single universal threshold a paddy farmer's CV of 0.8 is expected and not penalized in the same way it would be for an applicant claiming salaried employment.

C. Source-Wise Income Profiling

Income is treated as a portfolio of distinct streams rather than an undifferentiated total. The system identifies six primary income source types Salary, Remittance, Local Business/QR Collections, Freelance, Agricultural, and Cooperative/Interest and engineers features independently for each. Per-source metrics include monthly usable income (after reliability haircuts), recurrence ratio, volatility score, confidence score, and effective weighted contribution to EMI. This granularity allows the scoring model to distinguish between a borrower with stable salary income supplemented by modest freelance income and a borrower with volatile business income and no salary component, even if their aggregate monthly figures are identical.

D. Effective Monthly Income (EMI)

$$EMI = \sum_i (\text{Source Income}_i \times w_i) \quad (3)$$

where each weight w_i depends on source classification confidence, income stability, recurrence behavior, and documentary support. EMI is deliberately more conservative than raw observed inflow and serves as the foundation for all DTI and LTI calculations.

E. DTI and LTI Ratios

The Debt-to-Income Ratio estimates the proportion of effective monthly capacity consumed by total debt obligations (existing plus proposed):

$$DTI = \frac{\text{Existing Liabilities} + \text{Estimated New Loan Payment}}{EMI} \quad (4)$$

The Loan-to-Income Ratio measures structural credit exposure by comparing the requested loan amount to annualized EMI:

$$LTI = \frac{\text{Loan Amount}}{EMI \times 12} \quad (5)$$

Both ratios are computed in the Income Agent and passed to the Compliance Agent unchanged the Scoring Agent does not modify them, ensuring a clean separation between analytical recommendation and regulatory enforcement.

F. Tax Trust Score

The Tax Trust Score is a composite indicator derived from the presence of valid tax documentation, the consistency of the cited tax identification number across documents, and the effective tax rate:

$$\text{Effective Tax Rate} = \frac{\text{Tax Paid}}{\text{Declared Annual Income}} \quad (6)$$

Missing tax documentation reduces the Tax Trust Score but does not automatically disqualify informal borrowers it adjusts the scoring model's confidence in the declared income figure and may reduce the maximum approvable loan amount.

G. Asset-Backing Signal

Ownership documents such as *Lalpurja* (land title deeds) are transformed into an asset-backing signal rather than counted as income. The land area in square metres, combined with a localized valuation index, contributes to collateral confidence without distorting cash-flow estimates. This engineered separation between income evidence and asset evidence is important for both risk accuracy and regulatory explainability.

H. Identity and Document Consistency

Cross-document consistency features are engineered from the agreement between submitted documents on key identity attributes, particularly citizenship numbers. The system distinguishes between hard mismatches (genuine conflicts that trigger AML protocol) and low-confidence mismatches (likely OCR artifacts that route to manual verification). This distinction prevents false fraud flags on legitimate applications while preserving identity consistency as a meaningful trust signal.

VII. MODEL & SCORING APPROACH

The scoring engine uses XGBoost [6] a gradient-boosted decision tree framework as its primary model. XGBoost is chosen over deep learning alternatives for three reasons specific to this deployment context. First, it performs reliably on tabular data with mixed feature types (continuous ratios, binary flags, categorical income source labels), which is precisely the structure of the feature vector produced by the Income

and Document agents. Second, it produces stable, interpretable feature importance rankings that support SHAP-based explainability [5] without requiring post-hoc approximation. Third, it trains efficiently on relatively small datasets and generalizes well to distribution-shifted populations an important property for a system that will initially be trained on synthetic data and progressively refined on real loan outcome data.

The current prototype uses a synthetic training dataset calibrated to Nepal’s informal sector.

Data Access

The sample synthetic dataset is available for view at: https://drive.google.com/file/d/113xUa6rYpyPwa_7trZdZbcRgrkvmsPg6/view?usp=sharing

Important limitation: The surrogate default label is derived from DTI trajectories and income volatility patterns rather than observed loan repayment outcomes. Because EMI is both a training feature and the denominator of DTI (the primary label proxy), the model learns a feature-weighted approximation of the NRB’s DTI threshold rule rather than an independent empirical default predictor. Until approximately 2,000 real outcome labels are available across all income types, the model’s output should be interpreted as a *compliance-weighted creditworthiness heuristic*, not an empirical default probability. Institutions deploying the system in Phase 1 must use it as a decision-support tool supplementing human officer judgment, not as an autonomous decision maker.

To address the risk of leakage between EMI and the DTI-derived label, the training pipeline excludes DTI and LTI as direct features and uses only the underlying income distribution metrics (CV, recurrence ratio, source weights) alongside document trust features. DTI and LTI are computed at inference time by the Income Agent and passed directly to the Compliance Agent without entering the scoring model’s feature vector.

The model outputs a score between 0 and 1000, which is mapped to four risk tiers aligned with NRB loan classification standards: Low Risk (700–1000), Medium Risk (550–699), High Risk (400–549), and Very High Risk (below 400). These tiers determine both lending decisions and regulatory provisioning requirements, with higher-risk loans subject to stricter review or rejection.

The transition from synthetic to real training data is the system’s primary roadmap milestone. The architecture is designed to accommodate online learning as actual loan repayment outcomes accumulate, the model is retrained quarterly on a rolling dataset that blends synthetic priors with observed outcomes, with the synthetic proportion declining as real data grows. After approximately 2,000 loans with observed outcomes across all income types, the system can be retrained entirely on real data with validated predictive accuracy.

VIII. COMPLIANCE & EXPLAINABILITY

Compliance enforcement and decision explainability are not supplementary features in this system they are load-bearing

architectural components. A credit system operating in Nepal’s regulated banking environment must be able to demonstrate, for every decision, exactly which regulatory rule was applied, what data supported that application, and what the applicant can do in response. The system is designed to produce this documentation automatically.

A. NRB Directive 2080 Enforcement

The Compliance Agent enforces NRB Unified Directives 2080 as hard constraints evaluated in strict priority order [1]. The enforcement sequence is:

- 1) Hard identity mismatch $\Rightarrow$ REJECT (NRB AML Guidelines, KYC provisions)
- 2) Prohibited loan purpose $\Rightarrow$ REJECT (NRB Directive 2080, Clause 4.2)
- 3) Sector exposure ceiling exceeded $\Rightarrow$ REJECT or QUEUE (NRB Directive 2080, 25% of Tier 1 capital). *Fallback:* if the real-time sector exposure query to the core banking system is unavailable, the application is placed in a PENDING queue and re-evaluated when the data feed is restored. Applications are never auto-rejected due to a data feed failure alone.
- 4) Single-obligor concentration breach $\Rightarrow$ REJECT (NRB Directive 2080, exposure limit provisions)
- 5) $LTI > 5.0 \Rightarrow$ REJECT (NRB Directive 2080, Clause 5.1)
- 6) $DTI > 0.5 \Rightarrow$ REJECT (NRB Directive 2080, Clause 5.2)
- 7) $DTI > 0.4 \Rightarrow$ MODIFY (principal or tenure adjusted to bring $DTI \leq 0.40$)
- 8) $Credit\ score < 400$ with $DTI < 0.3 \Rightarrow$ MANUAL REVIEW
- 9) PSL-eligible borrower $\Rightarrow$ TAG and pass forward (mandatory portfolio allocation)
- 10) All constraints satisfied $\Rightarrow$ APPROVE

Every enforcement decision is logged with the specific directive clause cited, the calculated ratio that triggered the rule, the original and any modified terms, and a timestamp. This log is the system’s regulatory audit trail, producible in full on demand for NRB supervisory inspection.

B. SHAP-Based Explainability

SHAP values [5] are computed for every scored application, providing the marginal contribution of each feature to the final credit score. The three features with the largest absolute SHAP values are surfaced as top risk drivers in the decision output. These drivers appear in both the internal audit trail (for loan officers and regulators) and the customer-facing explanation (translated into plain language).

A typical rejection explanation might read: “*Your credit score was affected primarily by three factors: your declared income was significantly higher than what your transaction history supports (income consistency gap), your existing debt obligations represent 47% of your estimated monthly income (repayment capacity), and your income pattern showed significant month-to-month variation over the past year (income stability). To improve your eligibility, consider providing additional income documentation such as cooperative society*

records, or applying for a smaller loan amount that brings your repayment obligations below 40% of your monthly income.”

For approved applications, SHAP drivers are included in the loan file as supporting documentation for the credit decision providing the loan officer with a clear rationale for why the system recommended approval and which factors were most significant.

C. KYC and AML Integration

The system includes foundational Anti-Money Laundering (AML) controls within the automated pipeline. Two primary checks are currently enforced. First, cross-document identity validation ensures consistency of citizenship numbers across submitted documents; any discrepancy is treated as a hard AML flag and halts automated processing pending manual verification. Second, Politically Exposed Person (PEP) screening is performed by matching applicant names against a curated PEP database maintained by the Compliance Agent. Positive matches are escalated for mandatory senior officer review prior to decision issuance.

However, full AML compliance is not yet achieved in the current implementation. Suspicious Activity Reporting (SAR) to Nepal’s Financial Intelligence Unit (FIU-Nepal), along with screening against UN consolidated sanctions lists and OFAC databases, is put off to Phase 2 due to dependency on formal banking integrations.

This limitation has direct regulatory implications. Under NRB Anti-Money Laundering Guidelines 2020, licensed financial institutions are required to screen all borrowers against UN sanctions lists prior to establishing any credit facility. As the current system does not perform this check autonomously, its outputs cannot be treated as final lending decisions. Accordingly, all Phase 1 APPROVE outcomes must undergo mandatory review by a human compliance officer, who performs sanctions screening and final AML clearance prior to disbursement. The system’s approval output should therefore be interpreted as “*recommended for approval, subject to compliance clearance*”. Full end-to-end automation is contingent on the integration of FIU-Nepal reporting and sanctions screening mechanisms in Phase 2.

IX. AGENT ORCHESTRATION LOGIC

The system is orchestrated using LangGraph [10], a framework built on top of LangChain that models agent workflows as typed state graphs. LangGraph’s conditional edge routing maps directly to the decision logic required by the Compliance Agent allowing the pipeline to branch deterministically based on compliance outcomes without requiring custom routing code. The shared `AgentState` dictionary is a typed Python dataclass that is validated at every stage boundary, preventing malformed outputs from propagating downstream.

A. AgentState Schema

The `AgentState` carries all inter-agent communication. Key fields are listed in Table II.

TABLE II
AGENTSTATE SCHEMA — FIELDS, TYPES, AND PRODUCING AGENTS

Field	Type	Producing Agent
<code>applicant_id</code>	str	Entry
<code>documents_parsed</code>	dict	Document Agent
<code>doc_trust_score</code>	float [0, 1]	Document Agent
<code>identity_flags</code>	list[str]	Document Agent
<code>income_profile</code>	dict	Income Agent
<code>emi</code>	float (NPR)	Income Agent
<code>imr</code>	float	Income Agent
<code>dti</code>	float	Income Agent
<code>lti</code>	float	Income Agent
<code>credit_score</code>	int [0, 1000]	Score Agent
<code>risk_tier</code>	str	Score Agent
<code>shap_drivers</code>	list[dict]	Score Agent
<code>fraud_flags</code>	list[str]	Score Agent
<code>compliance_decision</code>	str	Compliance Agent
<code>modified_terms</code>	dict None	Compliance Agent
<code>nrb_clauses_cited</code>	list[str]	Compliance Agent
<code>psl_eligible</code>	bool	Compliance Agent
<code>final_decision</code>	str	Decision Agent
<code>customer_explanation</code>	str	Decision Agent
<code>audit_trail</code>	list[dict]	All agents

B. Conflict Resolution

The system’s authority hierarchy is explicit and non-negotiable: Compliance Agent decisions override all other agent recommendations. A credit score of 850 with excellent SHAP drivers does not override a DTI violation or a sector concentration breach. This hierarchy is encoded in LangGraph’s conditional edge logic, not in any agent’s internal reasoning ensuring that compliance enforcement is structurally guaranteed rather than behaviorally dependent.

Four representative conflict scenarios illustrate the hierarchy in practice. *First*, a high-scoring application (score: 800) where the Compliance Agent detects DTI = 48%: the application is modified, with principal reduced until DTI $\leq$ 40%, and the applicant receives a modified offer. *Second*, a low-scoring application (score: 350) where all NRB ratios are compliant: the case is escalated to manual review the high IMR (4.2) that drove the low score is a fraud signal that a human officer must evaluate in context. *Third*, a creditworthy application where sector concentration is at 24.8% of Tier 1 capital: the Compliance Agent rejects or queues the application regardless of score macro-prudential limits are absolute. *Fourth*, a citizenship mismatch between tax certificate and citizenship certificate: the application is immediately rejected regardless of income or score, with an AML flag requiring manual identity verification before the applicant can reapply. *Fifth scenario (dual thin-file)*, An applicant submits documents with a `doc_trust_score` below 0.30 and a transaction history covering fewer than three months, triggering both thin-document and thin-transaction flags simultaneously. This combination the most likely profile for a first-time rural borrower, the primary target population of this system cannot be scored with sufficient confidence for any automated pathway. The pipeline halts at the Score Agent without producing a credit score and routes directly to manual review with a

HIGH_PRIORITY flag. The officer review brief includes: (1) a list of supplementary documents that would increase `doc_trust_score` above 0.60 (cooperative passbook, utility payment history, guarantor letter), (2) the minimum transaction history window required to activate the standard scoring pathway (six months), and (3) whether the applicant qualifies for the agricultural fallback track based on declared occupation.

C. Fallback and Failure Handling

Every external API call and LLM inference request in the pipeline is wrapped in a retry handler with exponential backoff and a maximum of three attempts. If a component fails after exhausting retries, the pipeline does not halt: instead, it enters a degraded mode where the affected agent’s output fields are marked as unavailable and the application is routed to manual review with a system-generated note describing which data sources could not be obtained. This ensures that infrastructure failures do not produce silent incorrect decisions.

For the LLM classification fallback in the Income Agent, unresolved transactions are not simply dropped: they are tagged as `UNKNOWN_INCOME` and included in the income profile at zero weight, with a flag indicating the volume of unclassified inflows. A high volume of unclassified inflows reduces the Data Completeness Score and may lower the maximum automated approval threshold, ensuring that thin classification coverage translates to appropriate caution rather than silent underestimation.

X. IMPLEMENTATION PLAN

A. Technology Stack

The core pipeline is implemented in Python 3.11. Agent orchestration uses LangGraph 0.2 [10] with typed state channels. Document processing begins with Tesseract 5.0 (`nep+eng` language packs) for OCR, which handles both Devanagari script and Latin-script fields from scanned documents; the raw OCR output is then passed to a locally hosted language model for structured field extraction, eliminating any per-call API cost. Transaction categorization uses rule-based pattern matching for the first pass and Llama 3.3-70B as the LLM classifier for unresolved transactions; the model is served locally and chosen specifically to minimise per-application inference latency while maintaining sufficient classification accuracy for structured financial transaction descriptions. The scoring model is trained and served using XGBoost 2.0 [6] with SHAP 0.45 [5] for explainability. Because all inference components — OCR, LLM classification, and scoring — run on self-hosted infrastructure, there are no per-application API charges.

Data storage uses PostgreSQL for the `AgentState` audit log and application records, with row-level security policies ensuring that loan officers can only access applications within their authorized portfolio. All PII fields (citizenship numbers, declared income, transaction amounts) are encrypted at rest using AES-256. The API layer is FastAPI, exposing a single `/assess` endpoint that accepts a multipart form upload

(document images plus bank statement CSV) and returns a JSON decision payload synchronously for straightforward cases and asynchronously for manual review escalations.

B. Phased Rollout

Phase 1 (Months 1–3): Core pipeline development and synthetic data validation. Completing the Document Agent OCR pipeline with accuracy benchmarking on a sample of 100 real Nepali government document scans, implementing the full Income Agent with transaction categorization for all six income source types, integrating the XGBoost model with the synthetic training dataset, and deploying the Compliance Agent with all NRB Directive 2080 hard rules. End-to-end testing covers 500 synthetic applicant profiles across all income types including agricultural fallback cases.

Phase 2 (Months 4–6): Bank integration and pilot deployment. Onboarding a partner financial institution (cooperatively licensed bank or development bank with rural lending mandate), obtaining a sample of anonymized historical loan records to begin real-outcome model retraining, integrating the bank’s core banking system for real-time sector exposure queries, and deploying a loan officer review interface for manual review queue management. The pilot targets 200 real loan applications processed through the system in parallel with the existing manual process.

Phase 3 (Months 7–12): Production scaling and model improvement. Quarterly model retraining on accumulated real-outcome data, SAR/FIU integration for AML reporting, expansion of document coverage to include cooperative society passbooks and remittance certificates, and performance benchmarking against the bank’s existing scoring methodology on matched applicant samples.

XI. EXPECTED IMPACT

The most direct impact of this system is the extension of formal credit access to borrowers who are currently excluded not because of high default risk but because of inadequate assessment infrastructure. Nepal Rastra Bank data indicates that the agriculture sector which employs approximately 65% of the rural labor force receives less than 12% of formal commercial bank lending. A system that can reliably score agricultural borrowers using harvest-cycle behavioral data and cooperative records addresses a structural capital misallocation that current tools cannot.

Fraud reduction is the second significant impact dimension. The document-income consensus mechanism introduces a cross-validation step that existing bank processes which typically review submitted documents in isolation do not perform. In markets where forged income certificates are a documented fraud vector, the IMR provides a scalable automated screening mechanism that requires no manual investigation for consistent applications and automatically flags divergent ones. The expected false positive rate legitimate applications flagged for review is manageable at approximately 8–12% based on simulation testing, and those flagged applications proceed to human review rather than automatic rejection.

Processing efficiency is the third impact dimension. A standard loan application at a Nepali commercial bank currently requires 5–15 business days for assessment. The automated pipeline is designed to complete assessment for standard cases in under three minutes (see Section X-A for latency estimates and caveats) Manual review cases are targeted at resolution within 24 hours. This reduction in processing time is particularly significant for agricultural borrowers: a paddy farmer applying for a planting-season loan in Ashadh cannot wait two weeks for a decision.

At the portfolio level, if the system processes 200 applications per month the Phase 2 pilot target and achieves the expected 8–12% fraud flag rate for inconsistent applications, it would surface 16–24 applications per month for enhanced manual scrutiny that would otherwise have been processed without cross-validation. Based on simulation testing against the synthetic dataset, the expected false positive rate among flagged applications (legitimate borrowers incorrectly flagged) is approximately 2.6% at the baseline IMR thresholds, equivalent to fewer than one false fraud flag per month at pilot scale.

XII. LIMITATIONS & FUTURE WORK

This system has four acknowledged limitations that are important to state honestly.

Synthetic training data. The XGBoost model is trained on synthetic labels derived from feature-level heuristics and South Asian microfinance default literature rather than observed Nepali loan repayment outcomes. This means the model’s default predictions carry uncertainty that will only be resolved with real outcome data. The system is designed to accumulate and incorporate real outcomes through quarterly retraining, but until approximately 2,000 labeled loans are available across all income types, its predictions should be treated as informed estimates rather than empirically validated risk scores. Banks deploying this system in its initial phase should use it as a decision-support tool supplementing human judgment rather than as a fully autonomous decision maker.

Wallet API access. eSewa and Khalti do not currently offer public transaction-history APIs accessible to third-party credit applications. The current implementation uses bank statement uploads PDF or CSV exports available from any Nepali commercial bank as the primary transaction data source. Real-time API integration with wallet platforms is architecturally ready but depends on data-sharing agreements that must be established through formal fintech partnership channels. We recommend the participating bank initiate such discussions with eSewa Fonepay as part of the Phase 2 rollout.

Rural digital access barrier. The system currently requires digital document uploads, which assumes smartphone access and sufficient connectivity. In Nepal’s high-mountain and remote Terai regions, this assumption does not hold universally. A branch-assisted input pathway where a bank staff member scans and uploads documents on the applicant’s behalf is a necessary operational complement to the online application channel.

AML completeness. The current system implements basic cross-document identity validation and flags PEP matches for human review, but does not yet integrate with FIU-Nepal’s reporting infrastructure or OFAC/UN sanctions databases. This gap is a known limitation for full AML compliance and is explicitly prioritized in the development roadmap.

Future work will focus on five areas: real-outcome model retraining as pilot loan data accumulates; expansion of the agricultural fallback track to include satellite-derived crop yield data (NDVI indices from publicly available Sentinel-2 imagery) as an additional income proxy for subsistence farmers; integration of cooperative society data APIs; multilingual customer explanation generation in Nepali, Maithili, and Bhojपुरi for Terai-region applicants; and a federated learning architecture that allows multiple participating financial institutions to collaboratively improve the scoring model without sharing raw borrower data.

XIII. CONCLUSION

Nepal’s informal economy is not a problem to be solved it is a population to be served. The 63% of economically active Nepalis who earn their living outside the formal record are not high-risk borrowers by nature. They are simply invisible to the instruments that banks currently use to assess risk. The system presented in this paper is designed to make them visible, on honest terms, without compromising the risk discipline that institutional lending requires.

Document-income consensus is the system’s central mechanism: the recognition that declared income and observed transaction behavior should tell a consistent story in any legitimate application, and that consistency is itself a creditworthiness signal that no existing scoring system in Nepal’s market currently measures at scale. The Income Mismatch Ratio operationalizes this principle as a fraud detection feature and a confidence weight simultaneously welcoming to genuine informal earners whose documents and behavior align, appropriately cautious when they diverge.

The multi-agent architecture addresses the practical complexity of building such a system at banking-grade reliability. Five specialized agents each with a clearly defined scope, a typed interface, and an explicit authority level in the decision hierarchy collaborate through a shared, auditable state to produce decisions that are both analytically grounded and regulatorily enforceable. The Compliance Agent’s unconditional veto authority over all upstream recommendations is not a constraint on the system’s intelligence; it is the structural guarantee that makes the system trustworthy to deploy.

The limitations acknowledged in this paper are genuine and define a clear path forward. Three milestones gate the transition from prototype to production. First, a benchmark of real OCR accuracy across 500+ Nepali government document scans from diverse quality levels must be completed before Phase 1 deployment. Second, Phase 1 must be operated as a decision-support tool with mandatory human AML clearance, not as an autonomous decision maker, until FIU-Nepal integration is complete. Third, real outcome model retraining

requires approximately 2,000 labeled loans; given standard 12–24 month loan tenors, the Phase 3 timeline for autonomous scoring is 24–30 months from first disbursement, not 12.

The system’s measurable value proposition is specific: a three-minute assessment cycle for standard applications, a cross-validation fraud flag mechanism with a simulated false positive rate below 3%, a documented fallback pathway for the agricultural borrowers most likely to lack a digital footprint, and full NRB Directive 2080 audit traceability for every decision. These are not aspirational claims they are design specifications against which the system can and should be evaluated once real deployment data is available. Every day that Nepal’s informal workers cannot access formal credit is a day that capital stays misallocated. This system offers a concrete, measurable, and institutionally defensible step toward changing that.

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